310-3311 IMAGE PROCESSING



Lecture 2

Control and Fine-tune

(Image Generation)

How to tune Stable Diffusion

When we use stable diffusion from Automatic1111, we usually can set certain parameters which define how result turns out in the end.

Change checkpoint model

Prompt editing

Extra model

Parameter tuning

Manual re-touch

Prompt vs Creativity

Noise prediction including the prompt

VS

Noise prediction without the prompt

CFG: 0 CFG: 20

not follow the prompt ➔ strict to the prompt

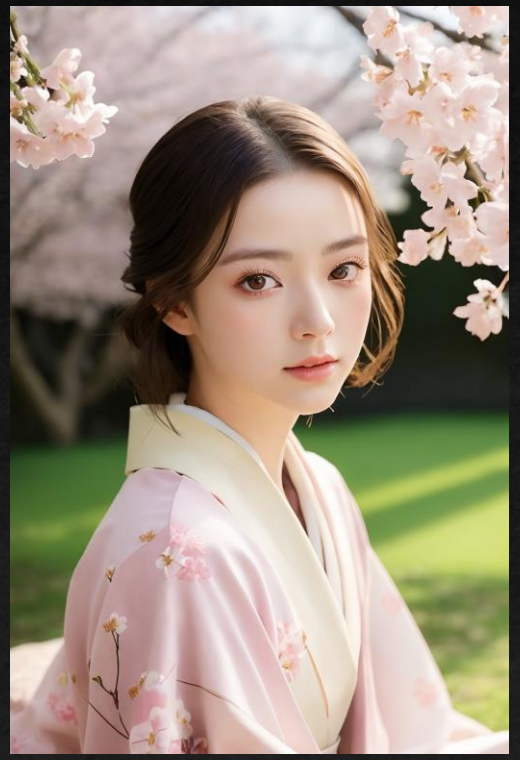
Classifier Free Guidance (CFG)



Classifier Free Guidance (CFG)

Random number (Seed)

seed parameter makes up the random number generator state



360



777

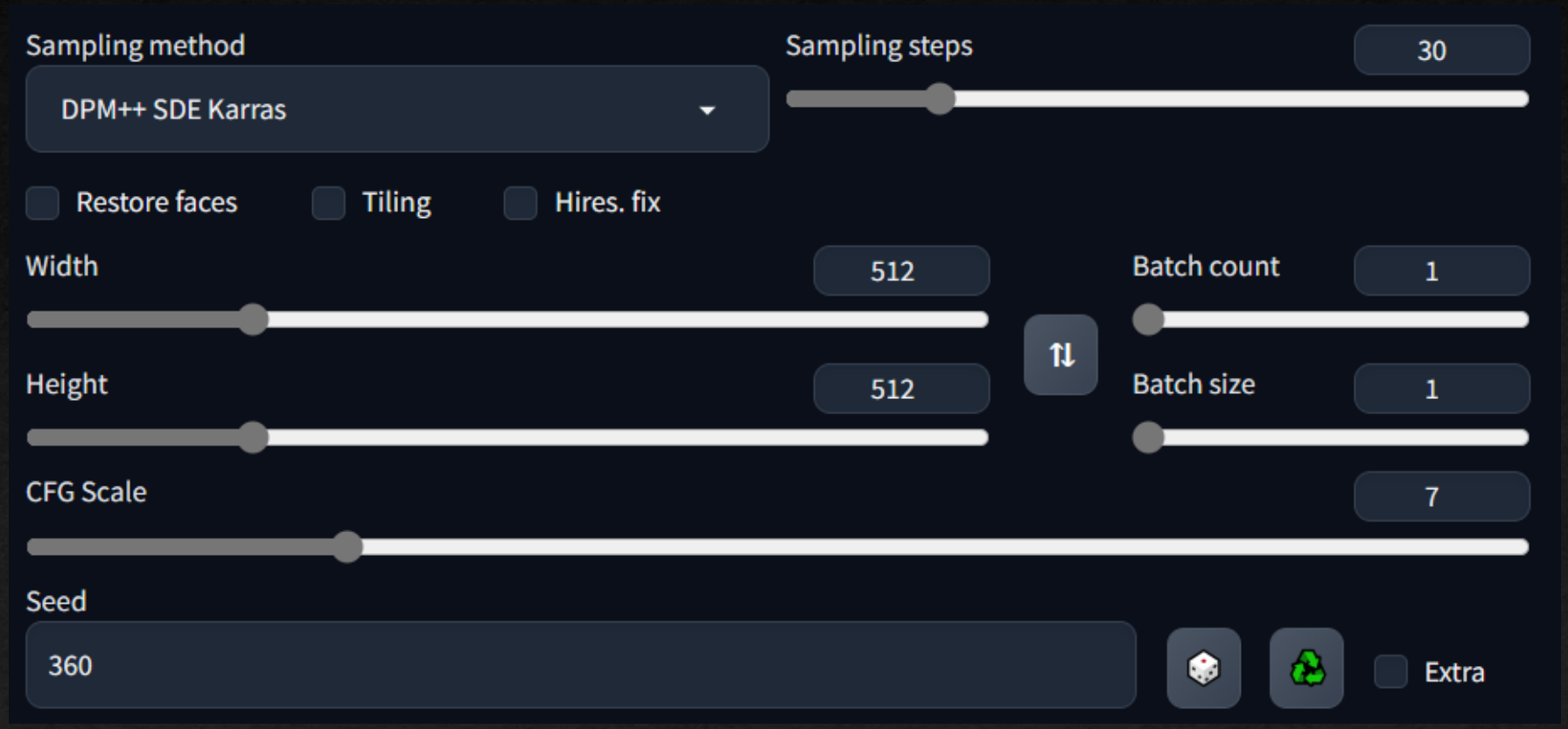


55

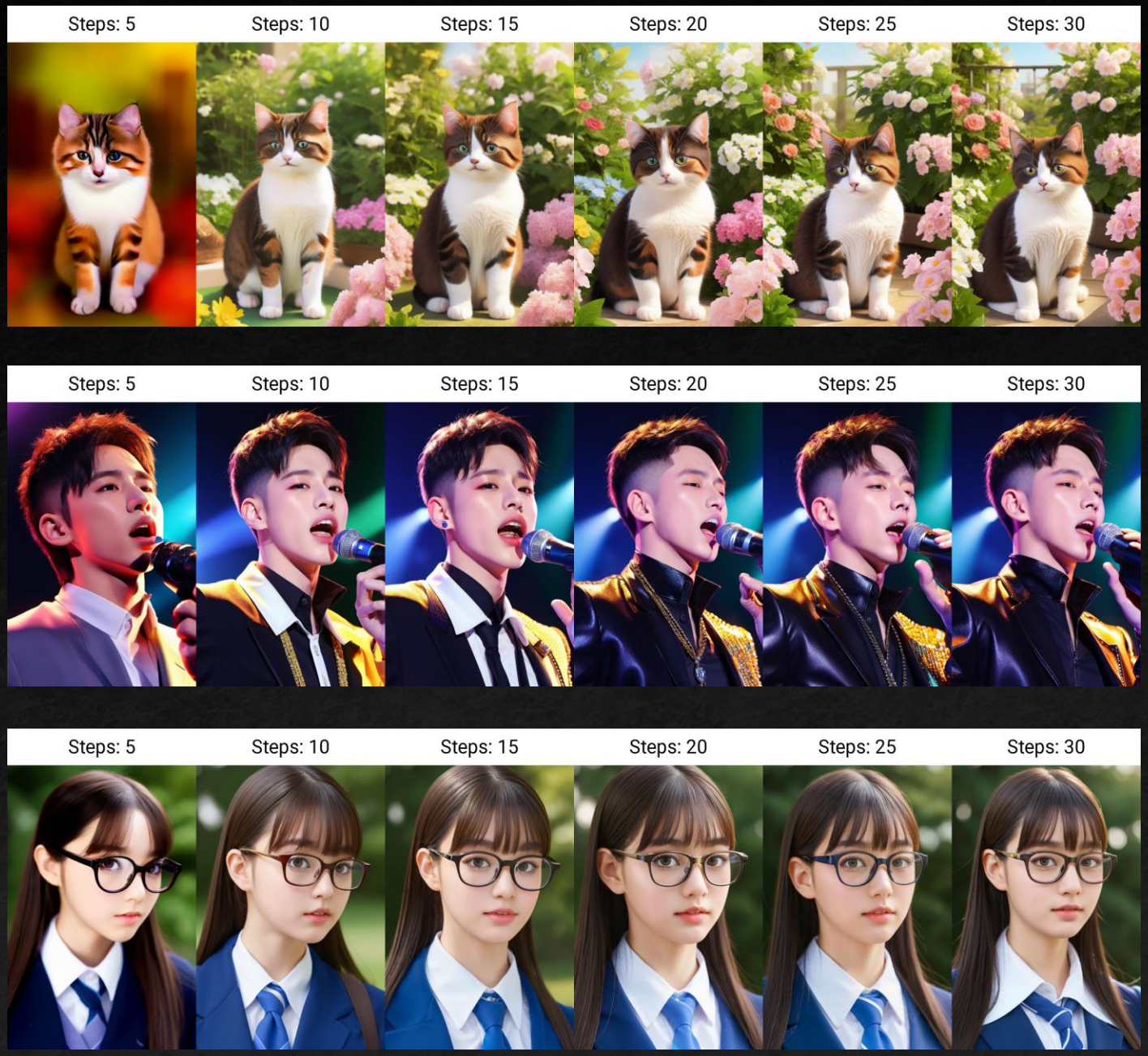
1girl, kimono,
sakura tree,

Random number (Seed)

seed parameter makes up the random number generator state



Seed: -1 (random)



Stable Diffusion Step

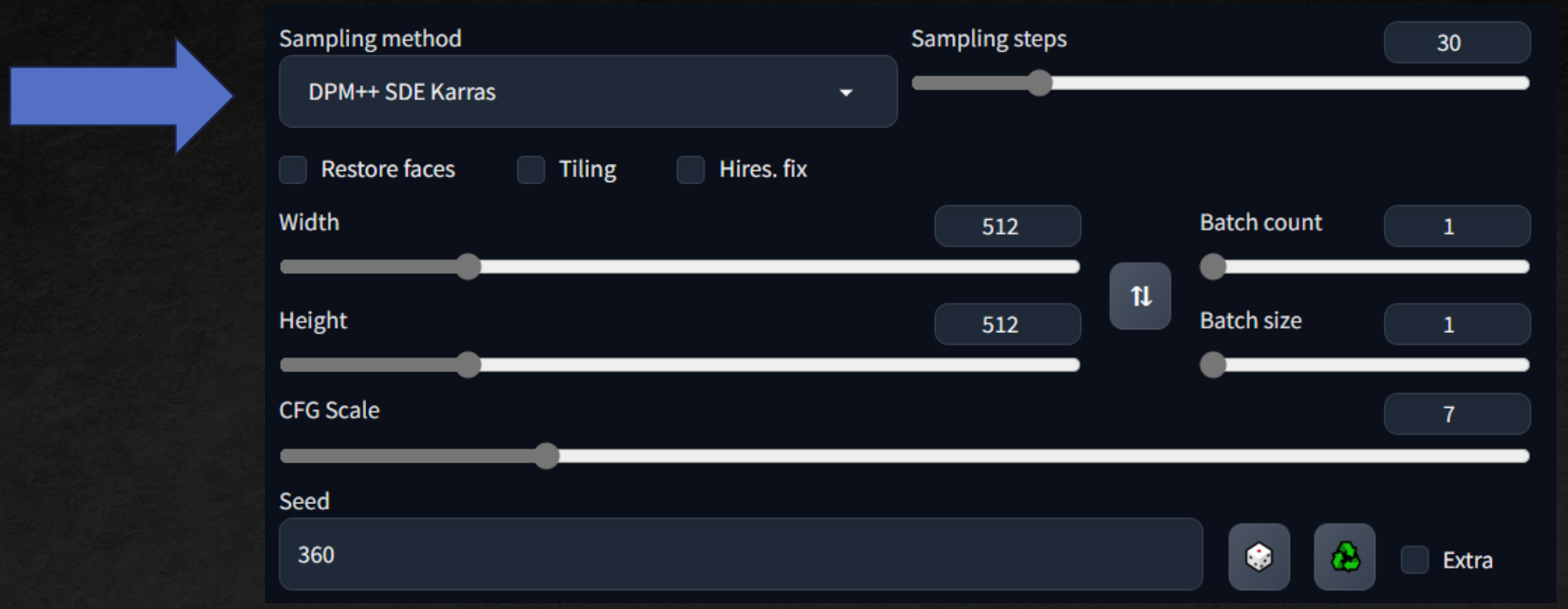
number of loop for denoising

more step = more compute = longer time

(usually around 20 – 60 steps is enough)

Sampler (method)

A rule that control how we denoising the image in each iteration



Sampler (method)



1	Prompt ✍🏽	→	Prompt generators + Lexica.art
2	Seed 🌱	→	Generate different images
3	CFG 🦮	→	Drive towards prompt, use 8-14
4	Sampler ✨	→	Noise schedule, DDIM for low step counts
5	Steps 🦶🏻	→	More steps ~ higher image quality, but 💰

Attention/emphasis

Attention/emphasis

Cheat sheet:

a (word) - increase attention to word by a factor of 1.1

a ((word)) - increase attention to word by a factor of 1.21 (= 1.1 * 1.1)

a [word] - decrease attention to word by a factor of 1.1

a (word:1.5) - increase attention to word by a factor of 1.5

a (word:0.25) - decrease attention to word by a factor of 4 (= 1 / 0.25)

X/Y/Z Plot

X/Y/Z Plot


multicolored
CFG Scale: 8.0
CFG Scale: 10.0
CFG Scale: 12.0
space
a castle
turquoise and gold
CFG Scale: 8.0
CFG Scale: 10.0
CFG Scale: 12.0
space
a castle

X/Y/Z Plot

Select which parameters should be shared by rows, columns and batch by using X type, Y type and Z Type fields, and input those parameters separated by comma into X/Y/Z values fields. For integer, and floating-point numbers, and ranges are supported.

Simple ranges:

- 1-5 = 1, 2, 3, 4, 5

Ranges with increment in bracket:

- 1-5 (+2) = 1, 3, 5
- 10-5 (-3) = 10, 7
- 1-3 (+0.5) = 1, 1.5, 2, 2.5, 3

Ranges with the count in square brackets:

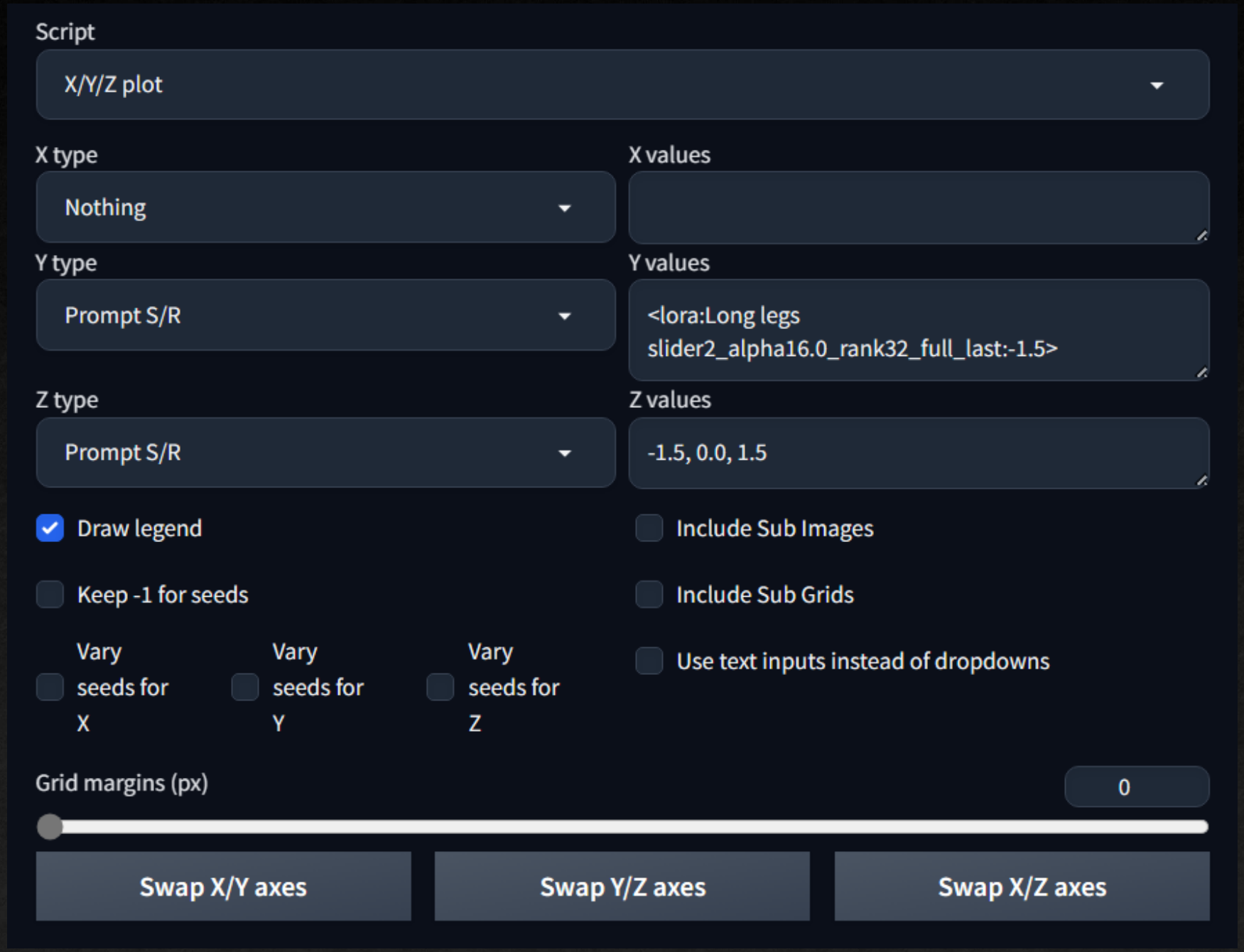
- 1-10 [5] = 1, 3, 5, 7, 10
- 0.0-1.0 [6] = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0

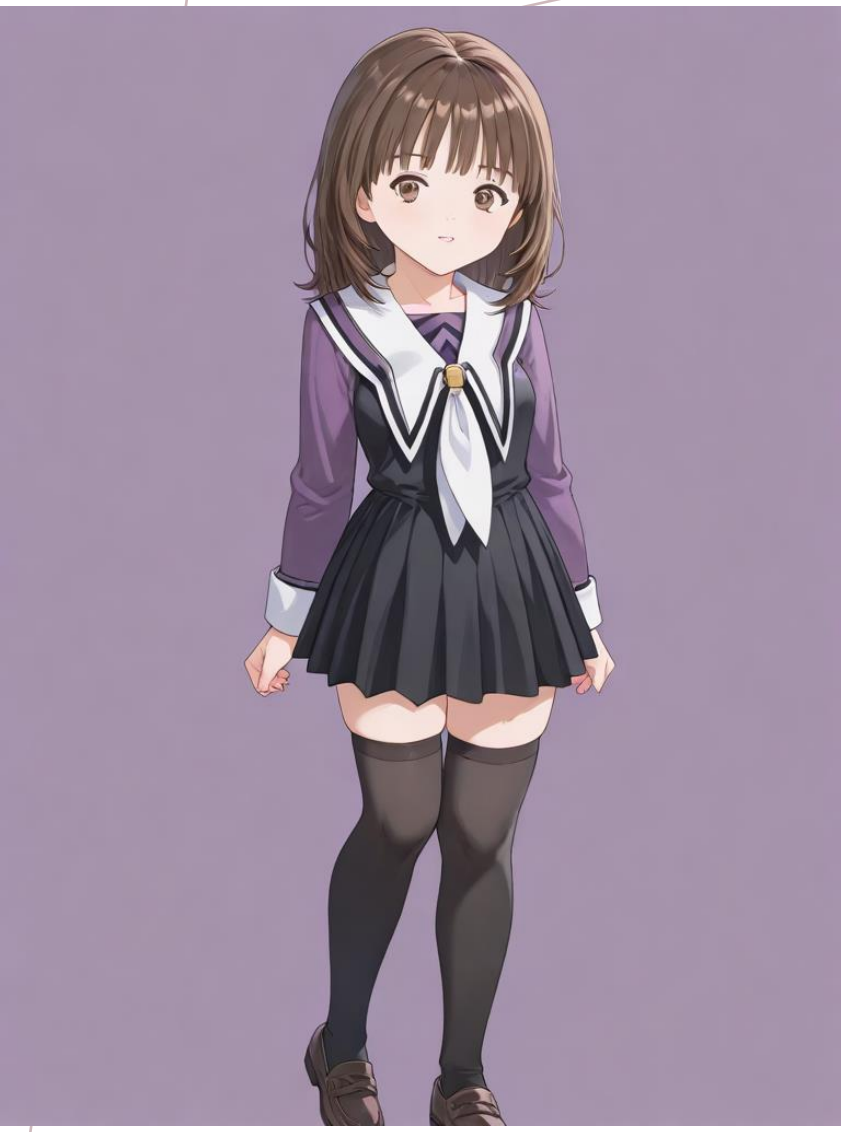
Prompt S/R

S/R stands for search/replace, and that's what it does - you input a list of words or phrases, it takes the first from the list and treats it as keyword and replaces all instances of that keyword with other entries from the list.

For example, with prompt a man holding an apple, 8k clean,
and Prompt S/R an apple, a watermelon, a gun you will get three prompts:

- a man holding an apple, 8k clean
- a man holding a watermelon, 8k clean
- a man holding a gun, 8k clean

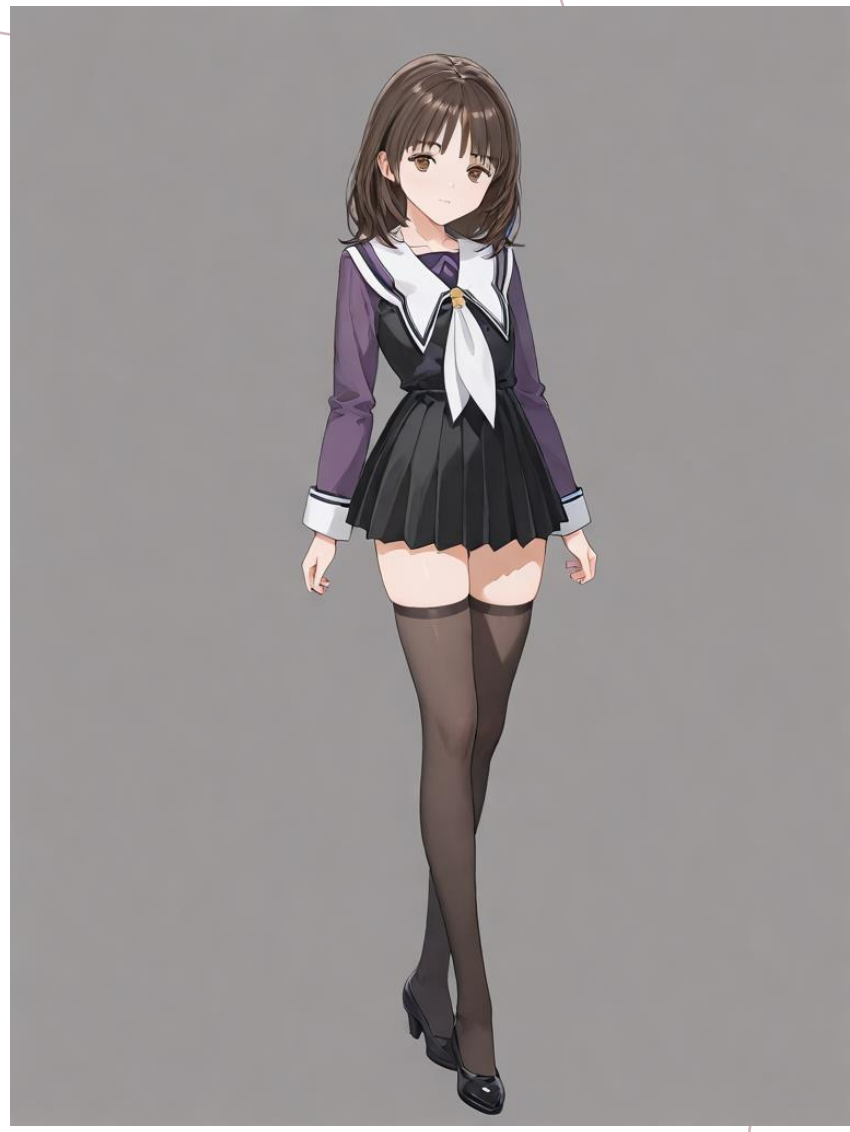

Script
X/Y/Z plot
X type
Nothing
X values
Y type
Prompt S/R
Y values
<lora:Long legs slider2_alpha16.0_rank32_full_last:-1.5>
Z type
Prompt S/R
Z values
-1.5, 0.0, 1.5
Draw legend
Include Sub Images
Keep -1 for seeds
Include Sub Grids
Vary seeds for X
Vary seeds for Y
Vary seeds for Z
Use text inputs instead of dropdowns
Grid margins (px)
0
Swap X/Y axes
Swap Y/Z axes
Swap X/Z axes



-1.5



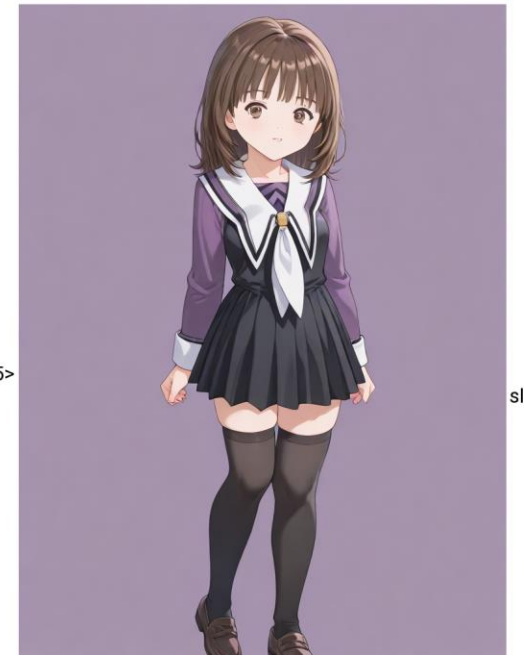
0.0



1.5

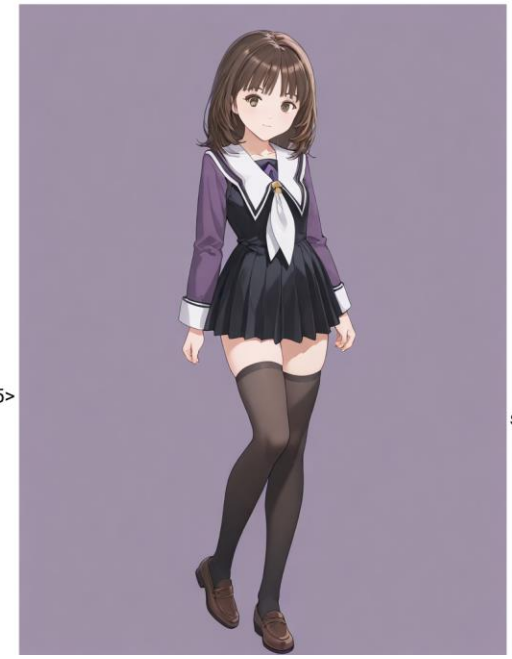
-1.5

<lora:Long legs slider2_alpha16.0_rank32_full_last:-1.5>



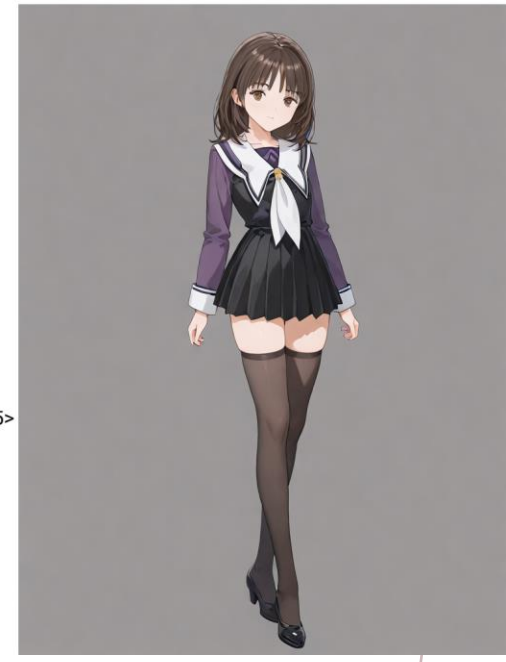
0.0

<lora:Long legs slider2_alpha16.0_rank32_full_last:-1.5>



1.5

<lora:Long legs slider2_alpha16.0_rank32_full_last:-1.5>



Workshop

ปรับอัตราส่วนของ Prompt ด้วย Emphasis และ/หรือ LoRA

แสดงผลลัพธ์โดยการใช้ X/Y/Z Plot

เปลี่ยน Seed ในแกน X ของ X/Y/Z Plot

และเปลี่ยน Prompt 1 ตัวในแกน Y ด้วย S/R Prompt

ControlNet & Regional Prompt

How to specific control the generation result

How to control the generation result

รูปภาพ 1 รูป แทนคำได้หลายคำ

ดังนั้นในทางกลับกัน

เราไม่อาจออธิบายรายละเอียดของภาพได้ด้วยเพียงคำบรรยาย

ControlNet

Why we need to use ControlNet ? ControlNet is a Stable Diffusion model that lets you copy compositions or human poses from a reference image.

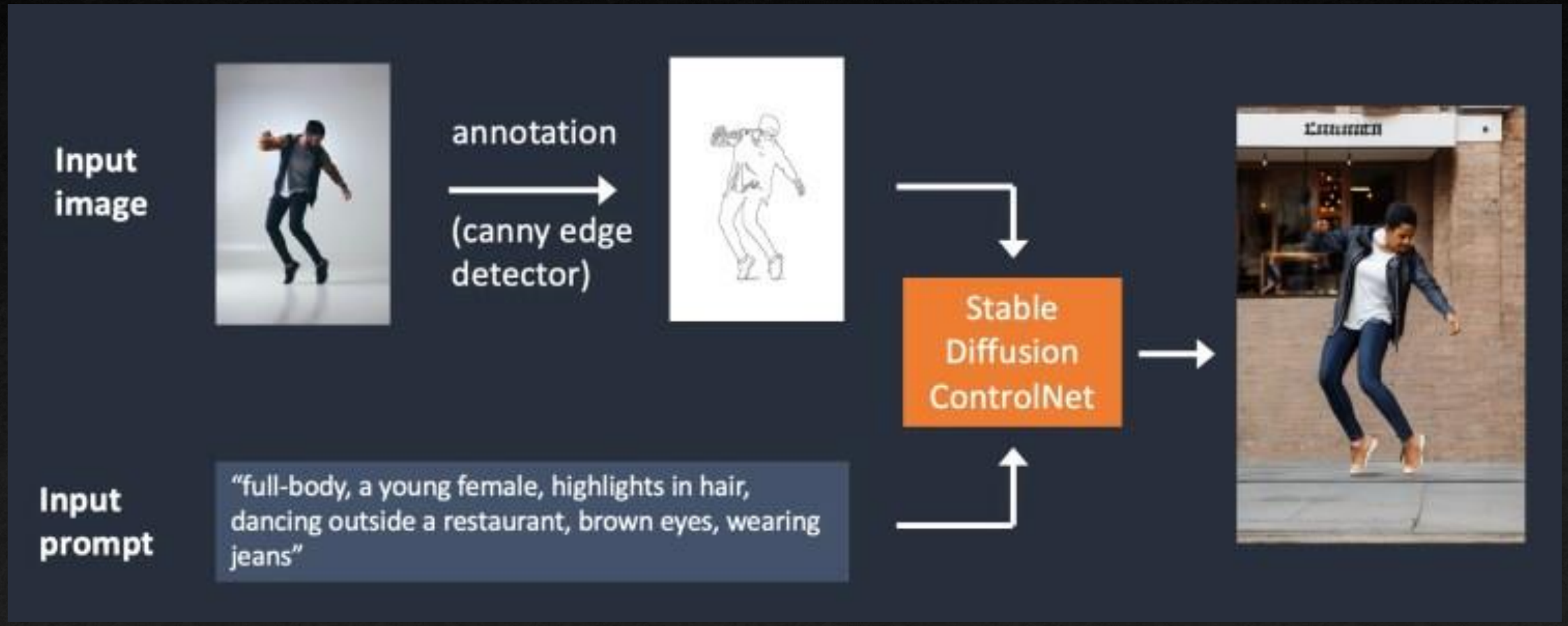
ControlNet is a neural network model for controlling Stable Diffusion models.

You can use ControlNet along with any Stable Diffusion models.

ControlNet adds one more conditioning in addition to the text prompt.

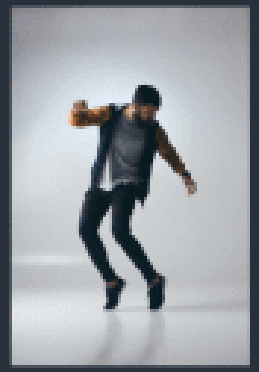
The extra conditioning can take many forms in ControlNet.

ControlNet


Input image
annotation
(canny edge detector)
Stable Diffusion ControlNet
Input prompt
"full-body, a young female, highlights in hair, dancing outside a restaurant, brown eyes, wearing jeans"

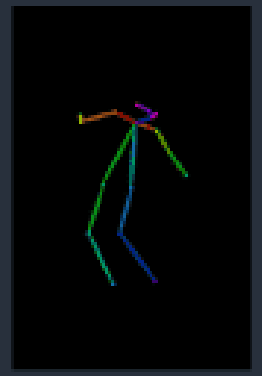
ControlNet

Input image



annotation

(openpose keypoint detection)



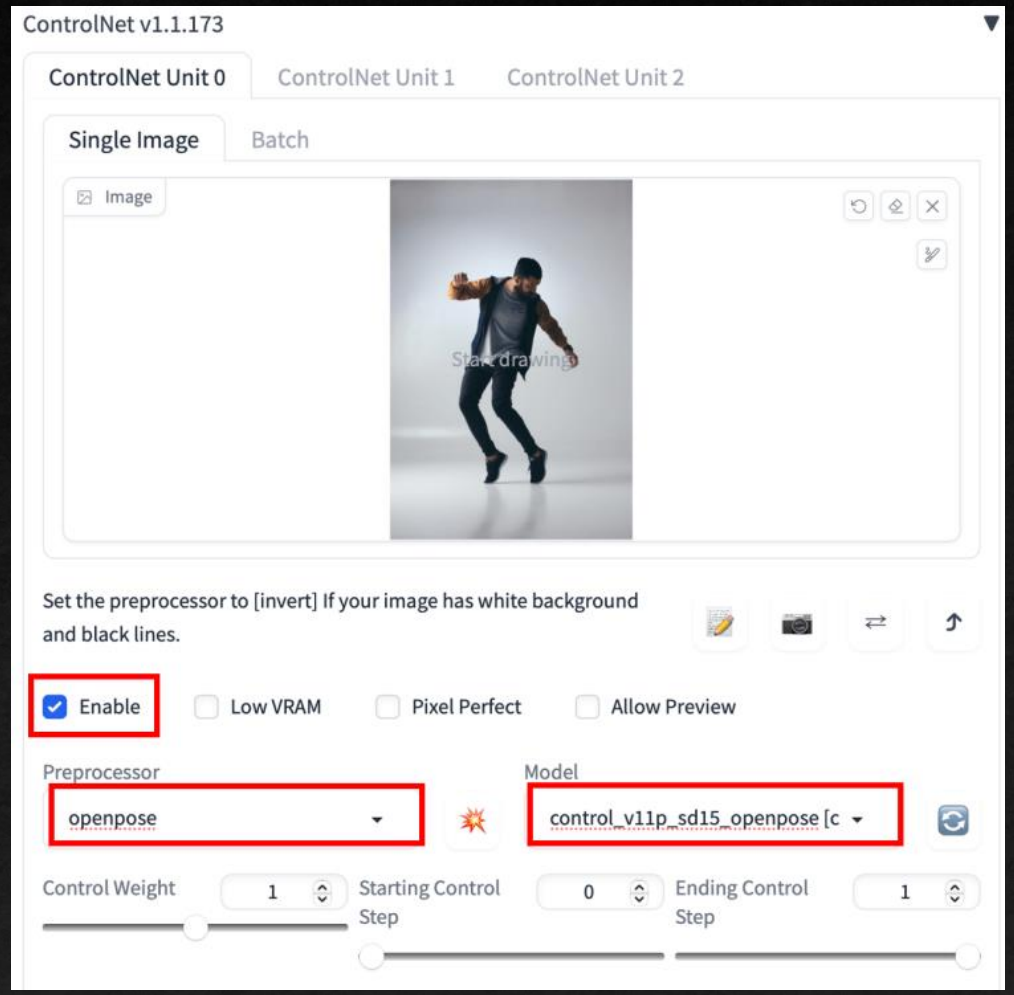
Stable Diffusion ControlNet



Input prompt

"full-body, a young female, highlights in hair, dancing outside a restaurant, brown eyes, wearing jeans"

Using ControlNet



- ❑ Download / Install ControlNet
- ❑ Upload an image to the image canvas
- ❑ Check the Enable checkbox
- ❑ Select a preprocessor and a model



Reference







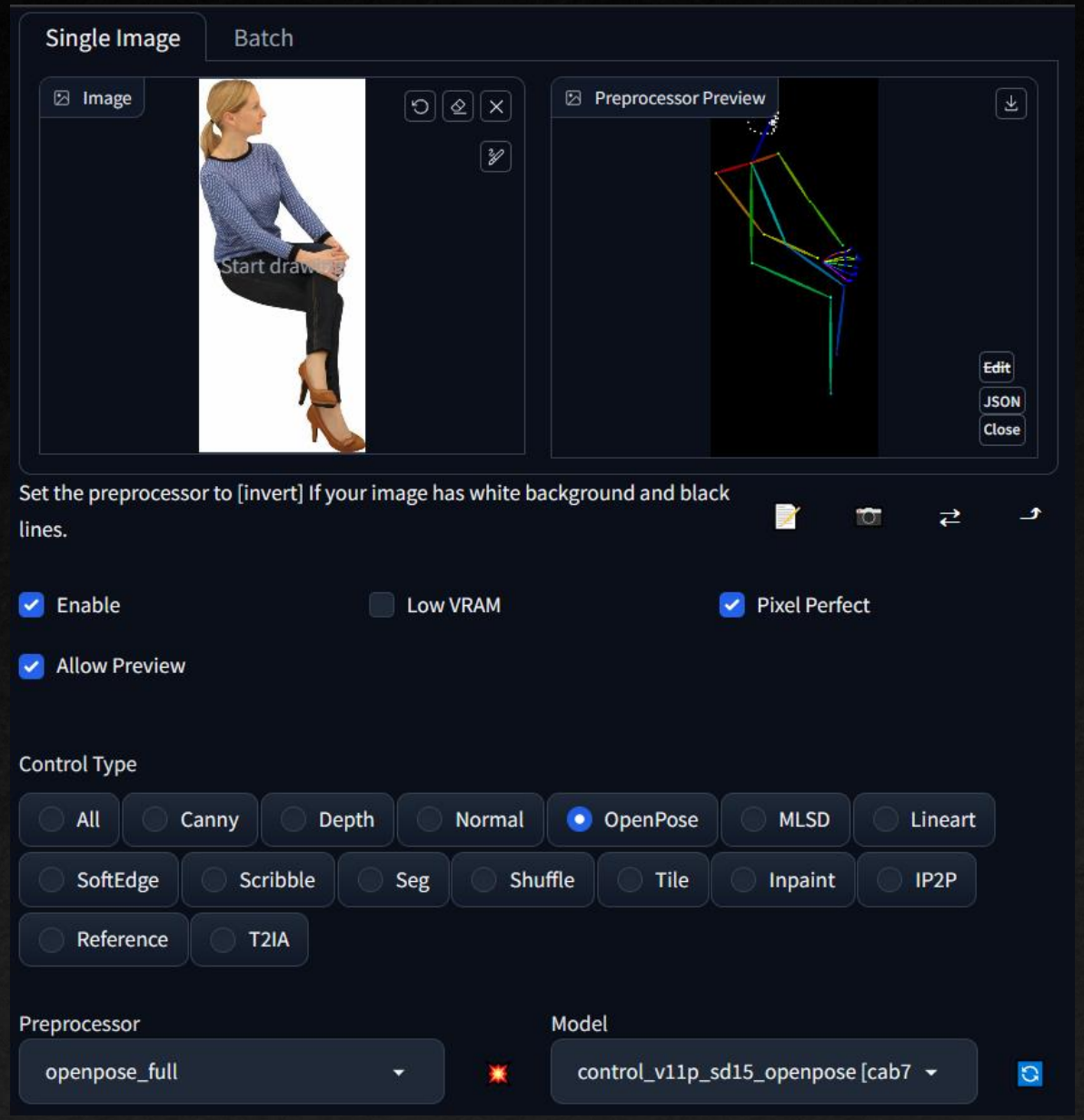


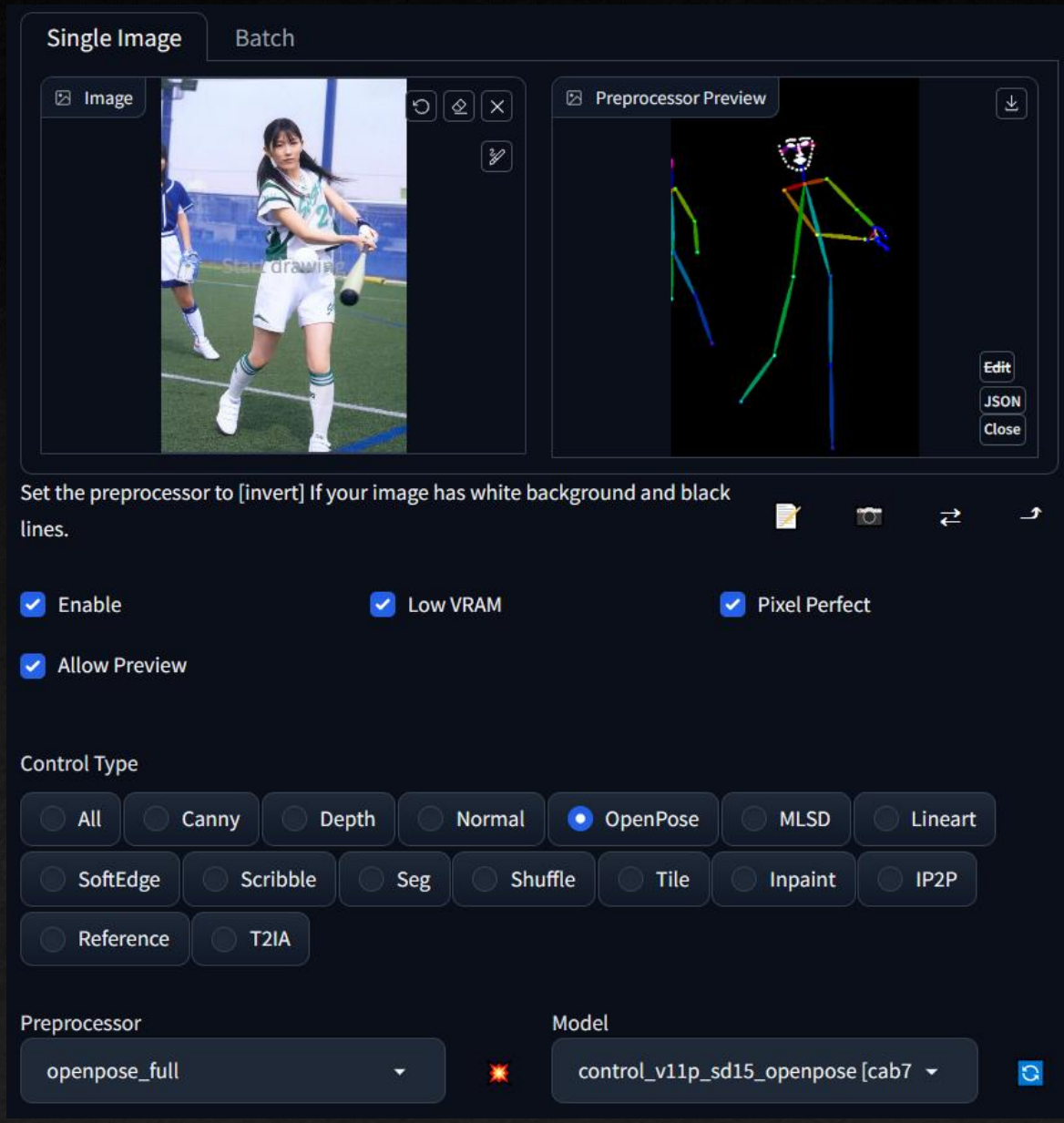
Example

Type of Preprocessor

The first step of using ControlNet is to choose a preprocessor. It is helpful to turn on the preview so that you know what the preprocessor is doing. Once the preprocessing is done, the original image is discarded, and only the preprocessed image will be used for ControlNet.

- ❑ Select **Allow Preview**.
- ❑ Optionally select **Pixel Perfect**. ControlNet will use the image height and width you specified in text-to-image to generate the preprocessed image.
- ❑ Click on the **explosion icon** next to the Preprocessor dropdown menu.


Single Image
Batch
Image
Start drawing
Preprocessor Preview
Edit
JSON
Close
Set the preprocessor to [invert] If your image has white background and black lines.
Enable
Low VRAM
Pixel Perfect
Allow Preview
Control Type
All
Canny
Depth
Normal
OpenPose
MLSD
Lineart
SoftEdge
Scribble
Seg
Shuffle
Tile
Inpaint
IP2P
Reference
T2IA
Preprocessor
openpose_full
Model
control_v11p_sd15_openpose [cab7


Single Image
Batch
Image
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Set the preprocessor to [invert] If your image has white background and black lines.
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T2IA
Preprocessor
openpose_full
Model
control_v11p_sd15_openpose [cab7



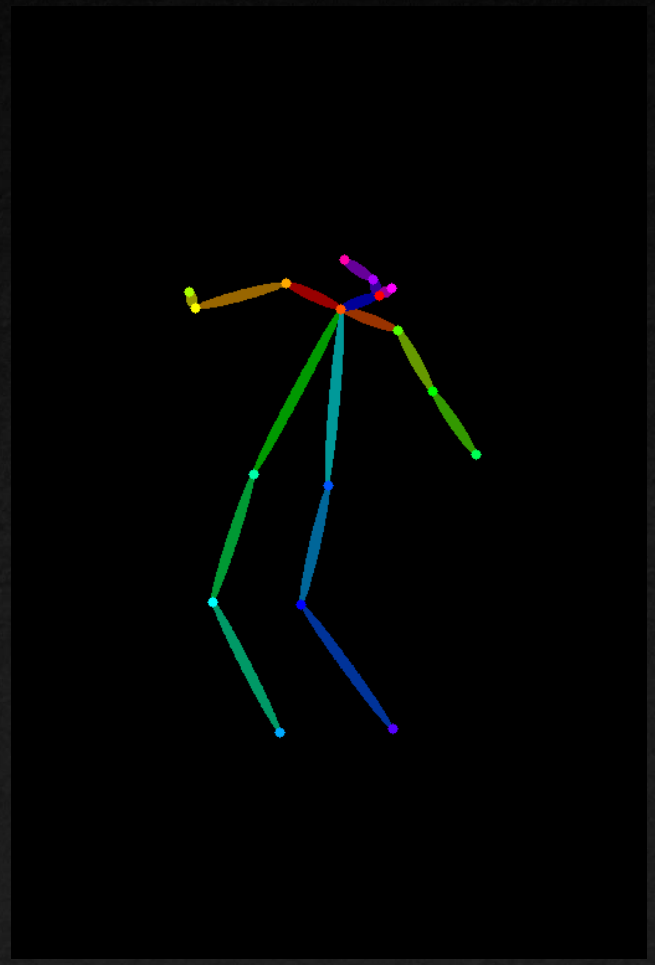






OpenPose



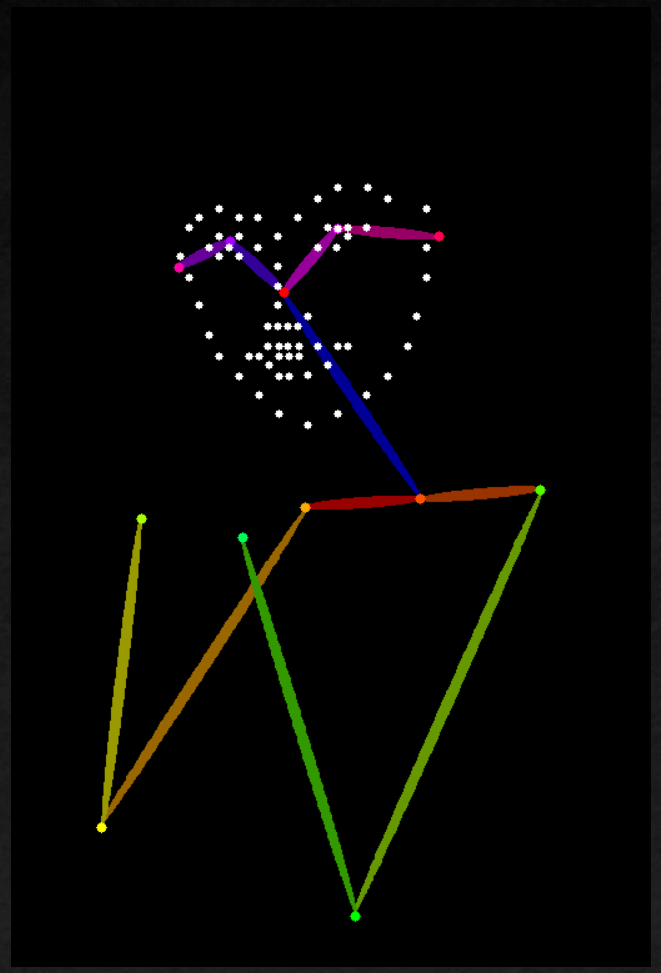


OpenPose:

eyes, nose, eyes, neck, shoulder, elbow, wrist, knees, and ankles.

OpenPose_face





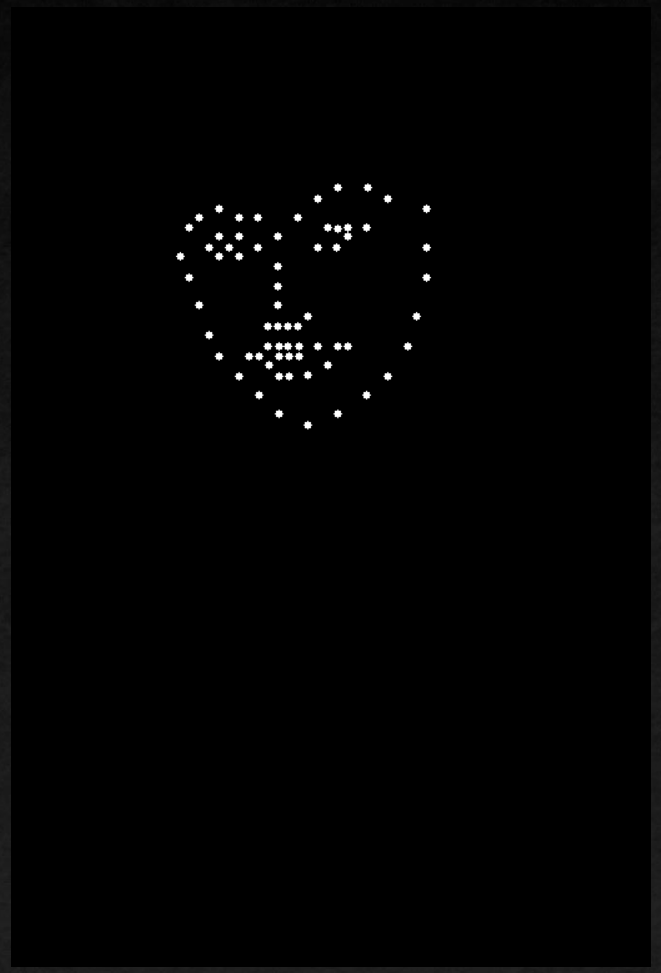
OpenPose (Face):

OpenPose_face does everything the OpenPose processor does but detects additional facial details.

It is useful for copying the facial expression.

OpenPose_faceonly



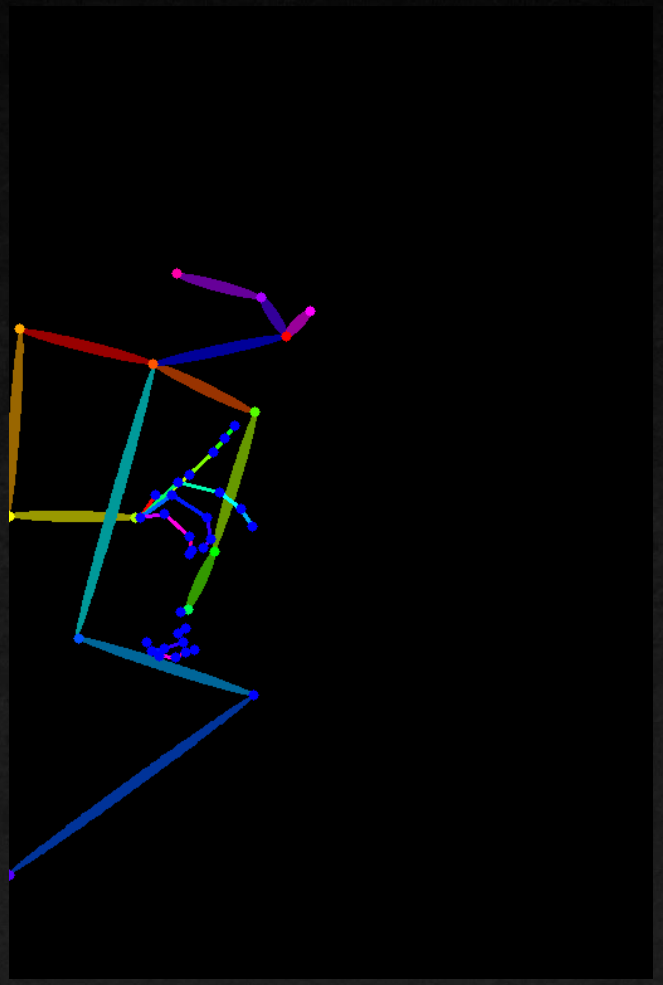


OpenPose (Face Only):

OpenPose face only detects only the face but not other keypoints. This is useful for copying the face only but not other keypoints.

OpenPose_hand



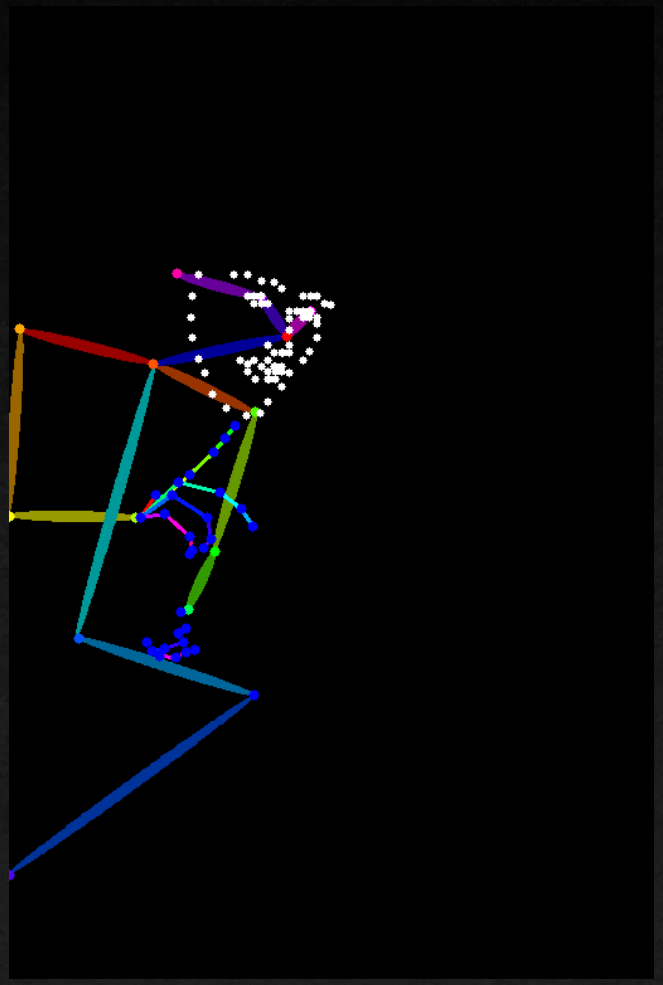


OpenPose (Hand):

OpenPose_hand detects the keypoint as OpenPose and the hands and fingers

OpenPose_full





OpenPose (Full):

OpenPose full detects everything openPose face and openPose hand do.

Canny (Edge detector)







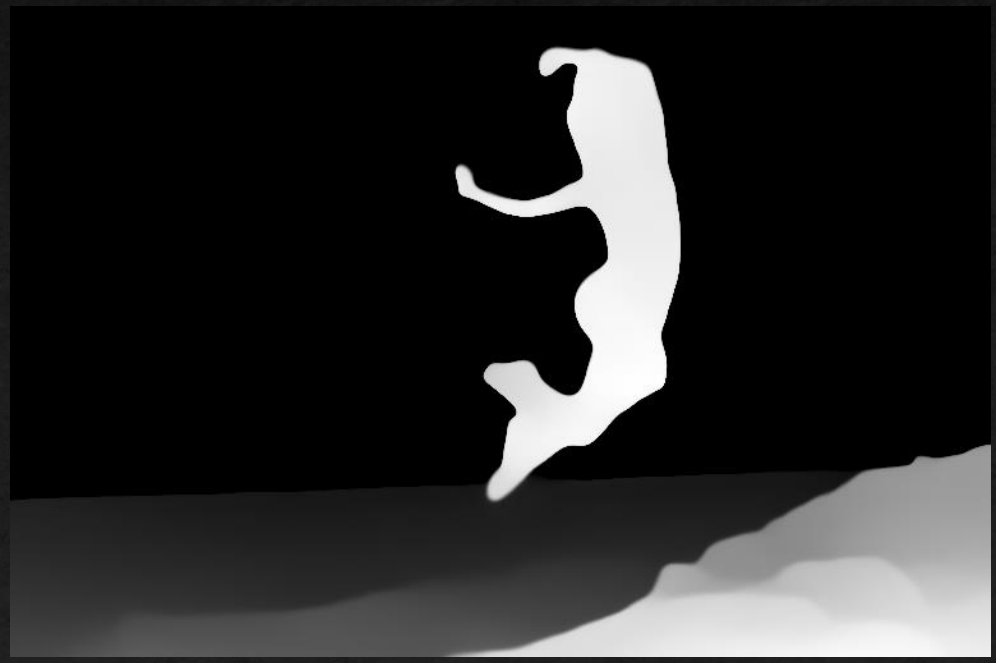


Canny edge detector is a general-purpose, old-school edge detector. It extracts the outlines of an image. It is useful for retaining the composition of the original image.

Depth



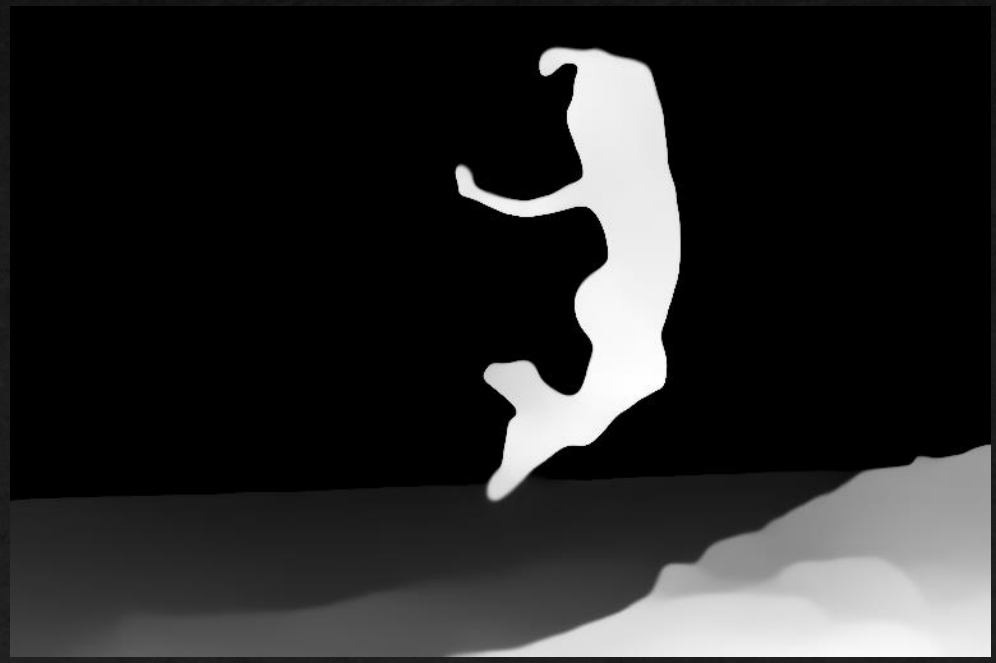
The depth preprocessor guesses the depth information from the reference image.



Depth



The depth preprocessor guesses the depth information from the reference image.

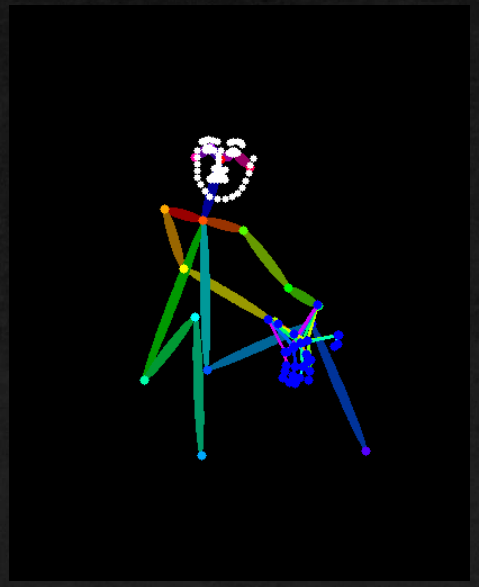


More than one ControlNet





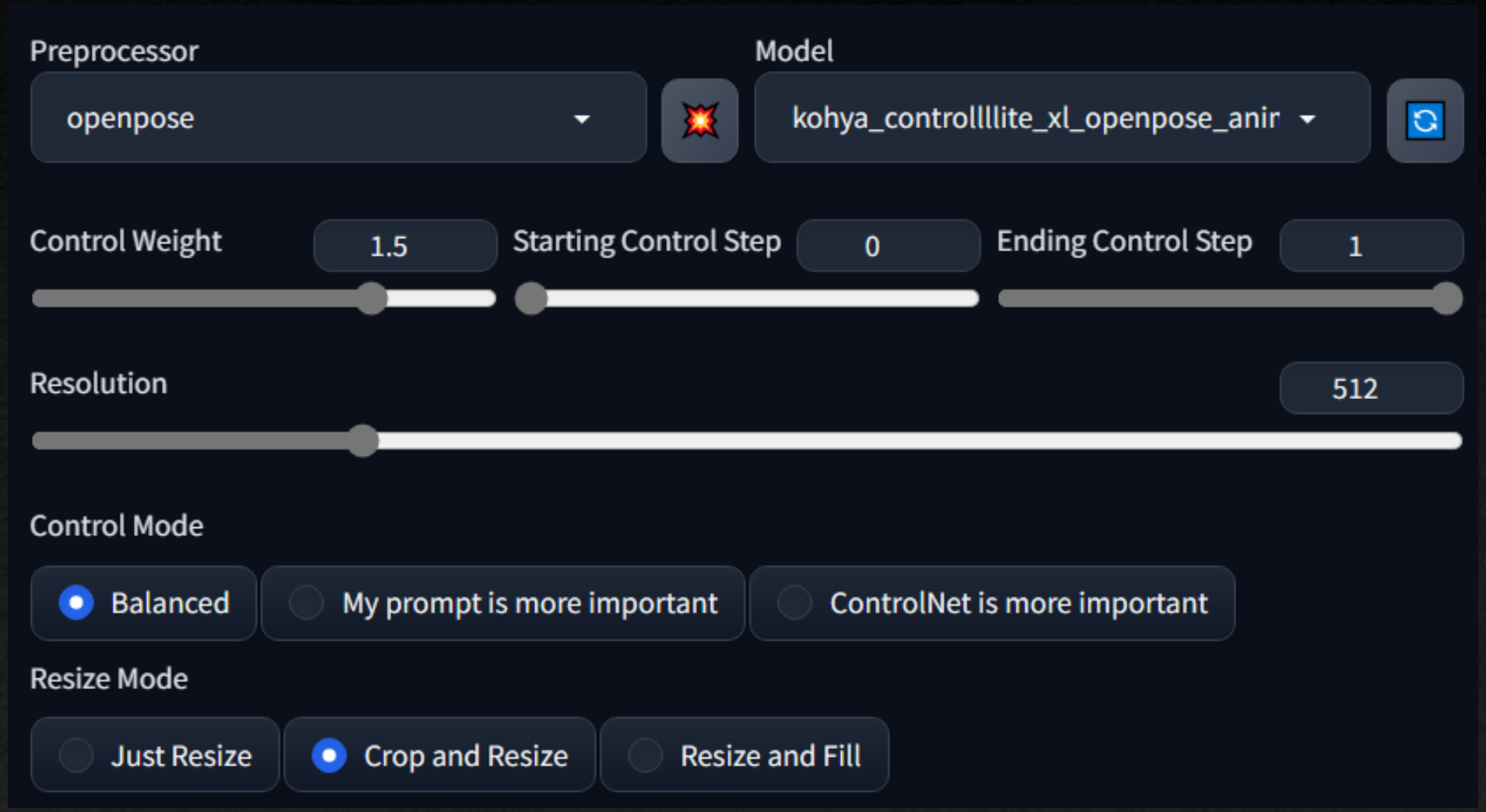




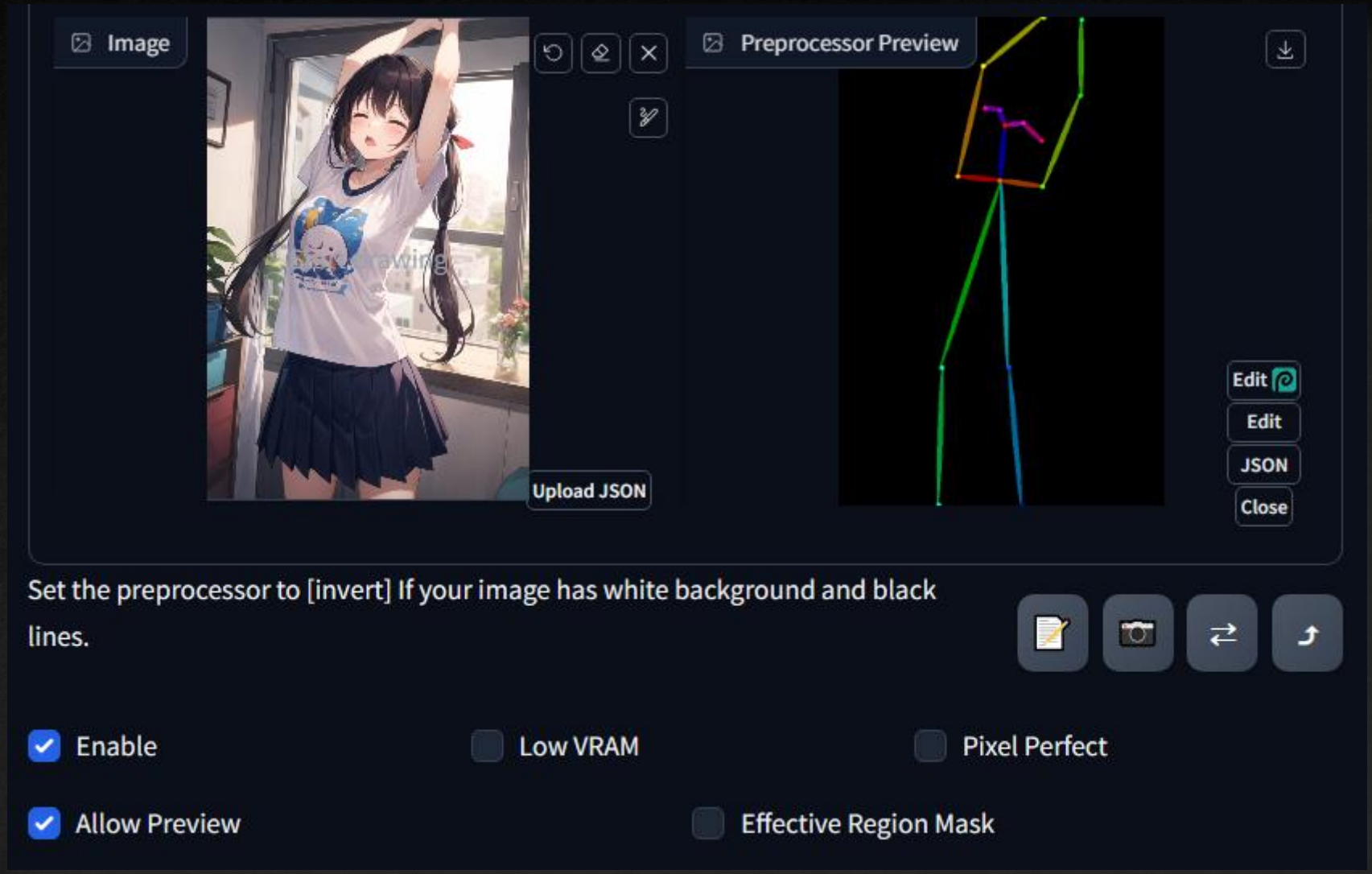


Do it....

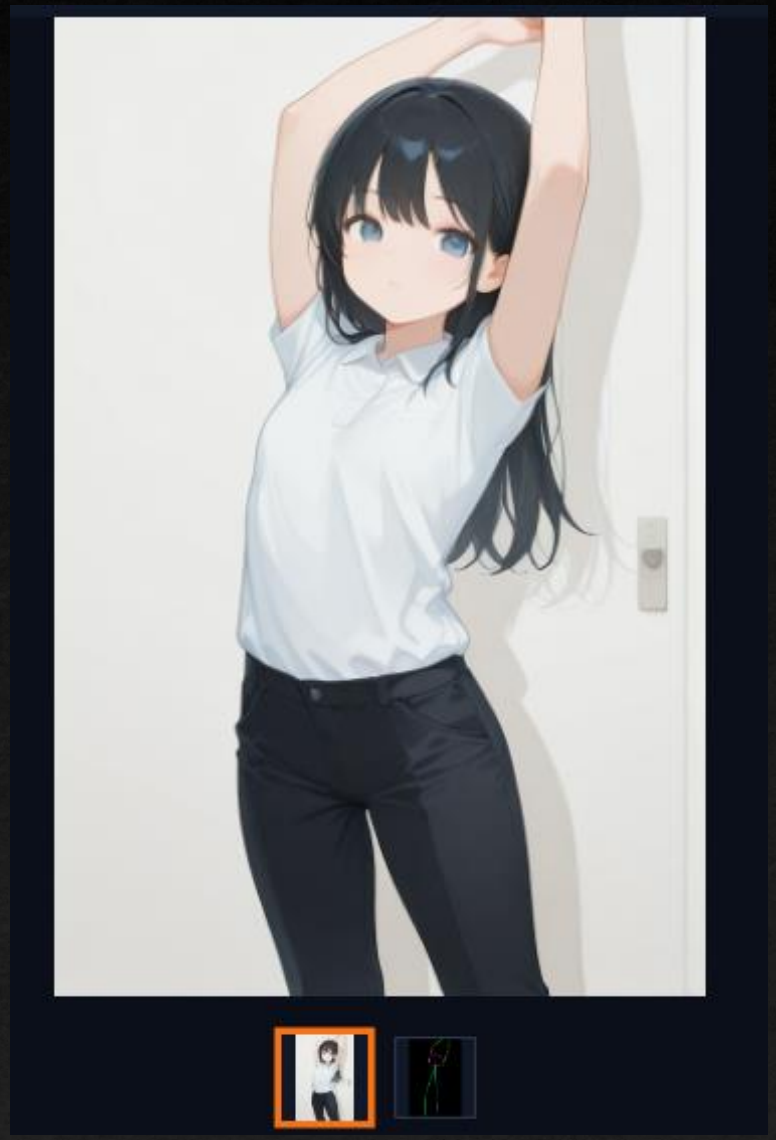
- ❖ เลือก Preprocessor และ Model ให้สอดคล้องกัน
- ❖ ปรับ Weight สำหรับ Control

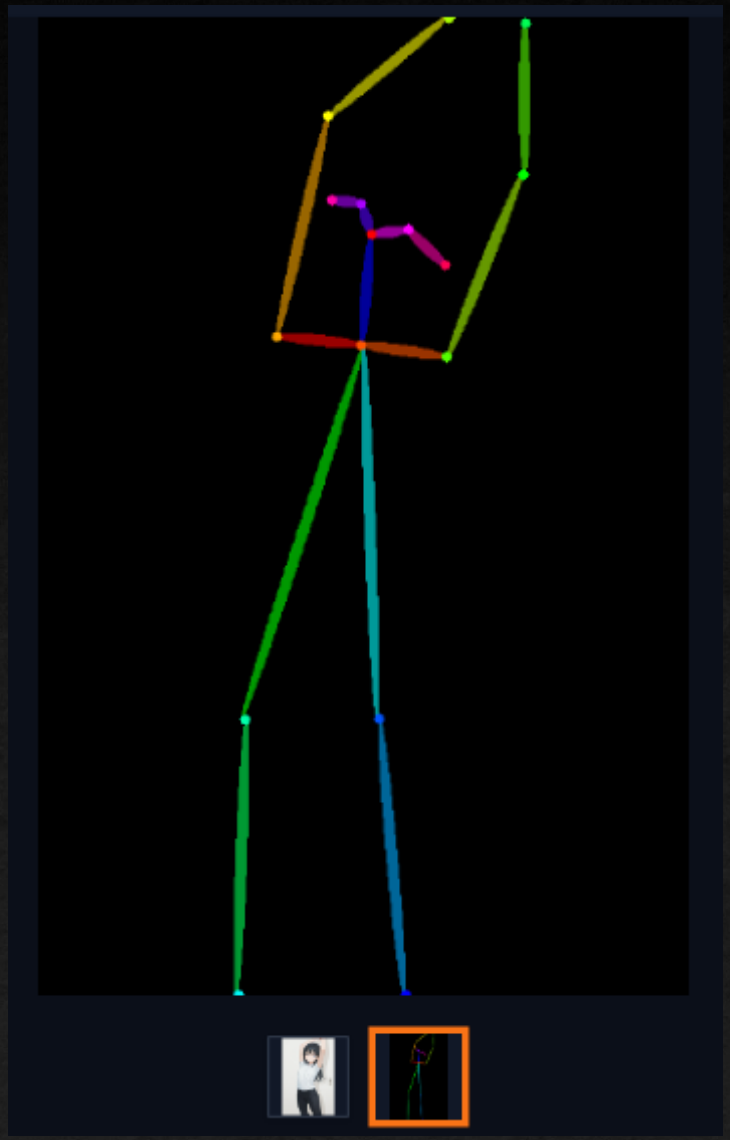


- นำภาพต้นแบบมาใส่
- Preview ผลลัพธ์



❖ ทดสอบผลลัพธ์ที่ได้ (*** ประสิทธิภาพ ขึ้นอยู่กับ Checkpoint + ControlNet model)





Regional Prompt **(Forge Couple)**

Why we need regional prompt? Let’s say you want to generate a man and a woman in the same image. Using the simple prompt: **a man and a woman**





Regional Prompt

But what if you want to be more specific? Like generating a man with black hair and a woman with blonde hair? Naturally, you write that in the prompt:

a man with black hair, a woman with blonde hair









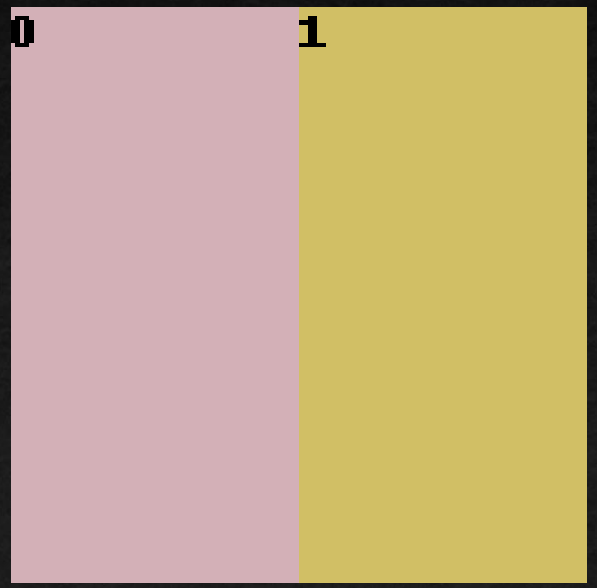
Regional Prompt

- Unfold the Regional Prompter section on the txt2img page.
- Check Active to activate the regional prompter.
- Set parameter
  - Divide mode: Horizontal
  - Generation mode: Attention
  - Divide Ratio: 1, 1
- Click **visualize and make template**.
- Put prompt:

  a man and a woman, a man with black hair

  BREAK

  a man and a woman, a woman with blonde hair



Regional Prompt



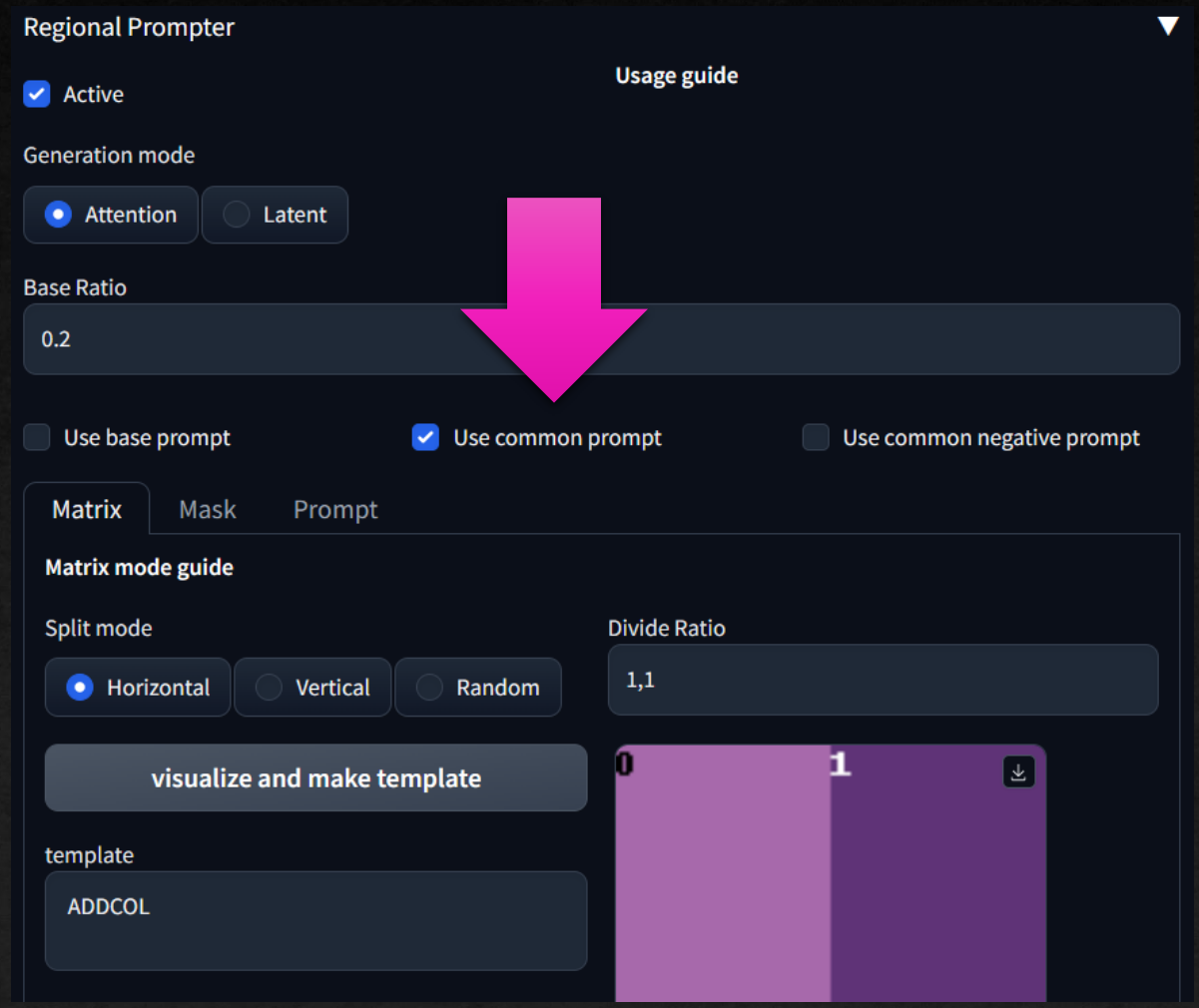






Regional Prompt

Tips



a man and a woman

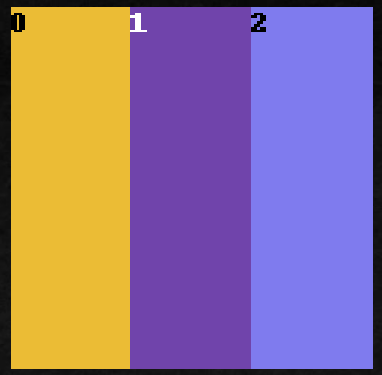
BREAK

a man with black hair

BREAK

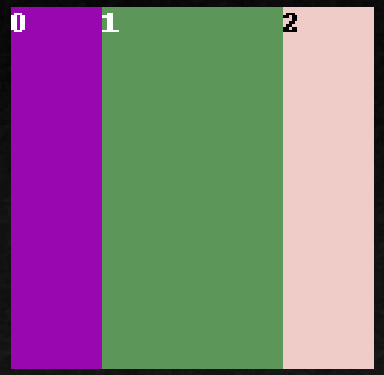
a woman with blonde hair

Regional Prompt



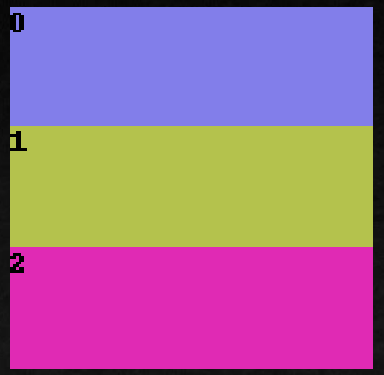
Horizon

(1,1,1)



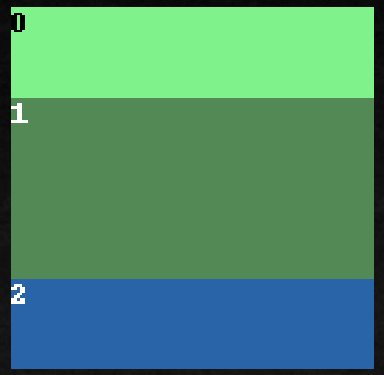
Horizon

(1,2,1)



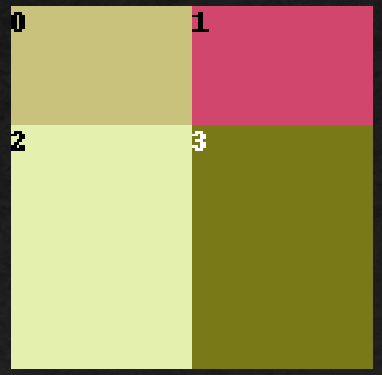
Vertical

(1,1,1)



Vertical

(1,2,1)

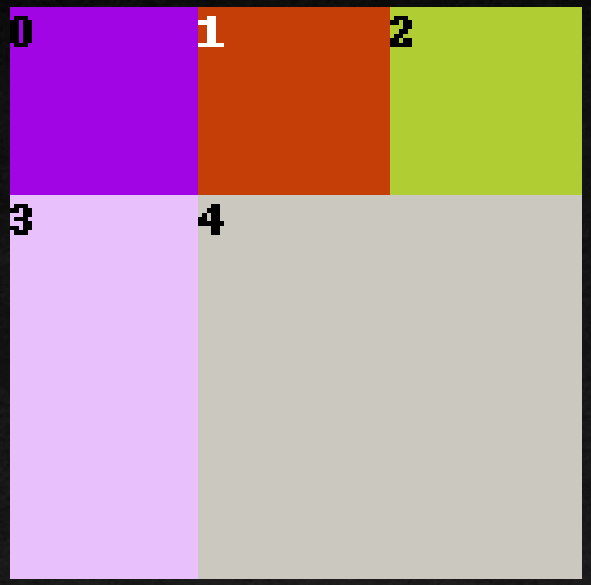


Horizon

(1,1,1; 2,1,1)

The **first number** in each row represents the **height** of the row.
The subsequent numbers represent the **width of the regions**.

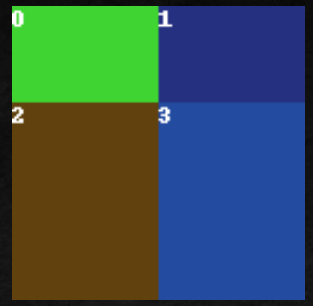
Regional Prompt



Horizon

(1,1,1,1; 2,1,2)

Example



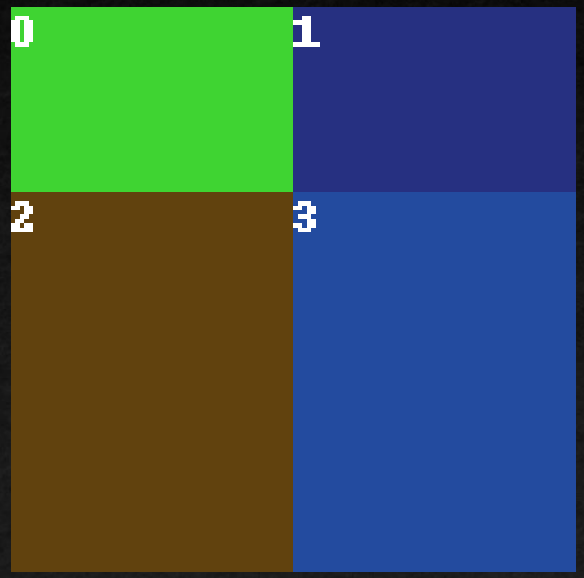
Horizon

(1,1,1;2,1,1)

a witch, highly detailed face, half body, studio lighting, dramatic lighting, highly detailed clothing, looking at you, mysterious, dramatic lighting
BREAK
(full moon:1.3)
BREAK
BREAK
BREAK
(beautiful fire magic: 1.2)

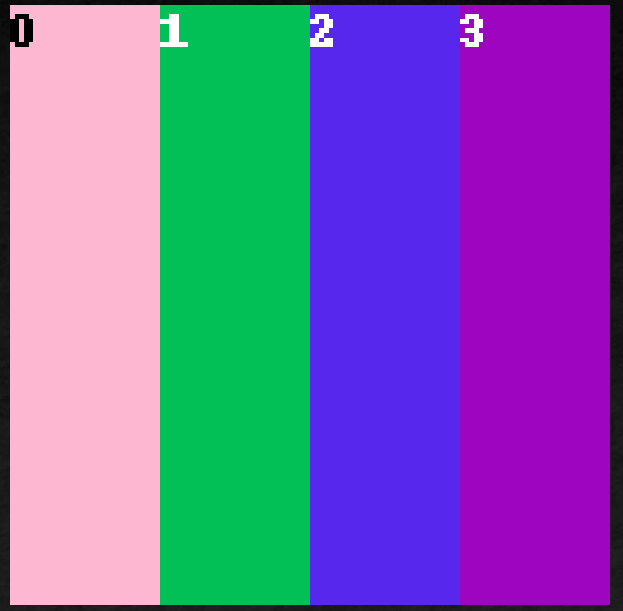
Regional Prompt





Regional Prompt





Do it....

Base Ratio

0.2

Use base prompt | Use common prompt | Use common negative prompt

Matrix | Mask | Prompt

Matrix mode guide

Main Splitting

Columns | Rows | Random

Divide Ratio

1,1

Width 1024

Height 1024

visualize and make template

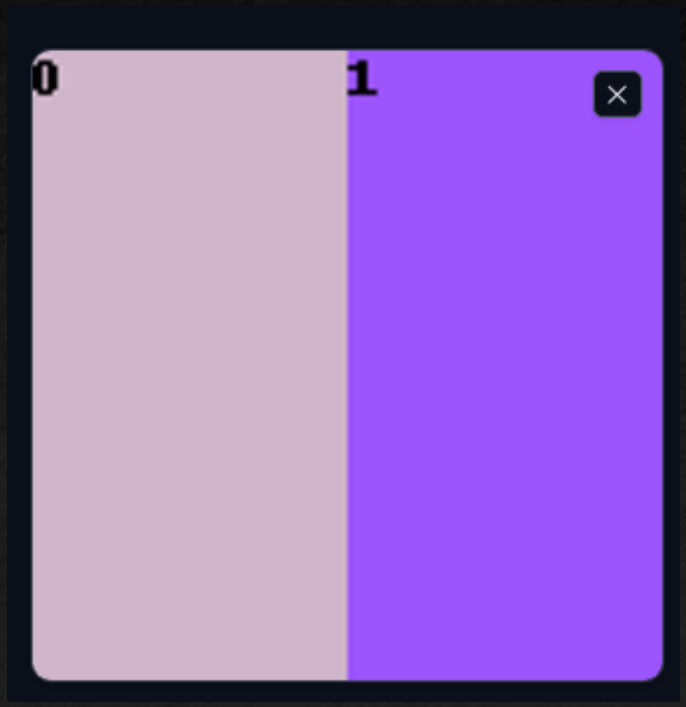
template

ADDCOMM
ADDCOL

flip "," and ";"

Overlay Ratio 0.1

- เลือก [Use common prompt] → ภาพรวม
- ปรับขนาดภาพให้ถูกต้อง
- เลือก Overlay Ratio ตามที่ต้องการ
- Visualize and make template



```
2girls, white background,
BREAK
1girl, blonde hair, medium hair,  hair, black shirt, denim shorts,
BREAK
1girl, black hair, long hair, white tank top, pink dolphin shorts,
```

- เขียน Prompt ตาม Template
- แบ่งวรรคด้วย <Keyword>
- สามารถใช้ BREAK แทนได้



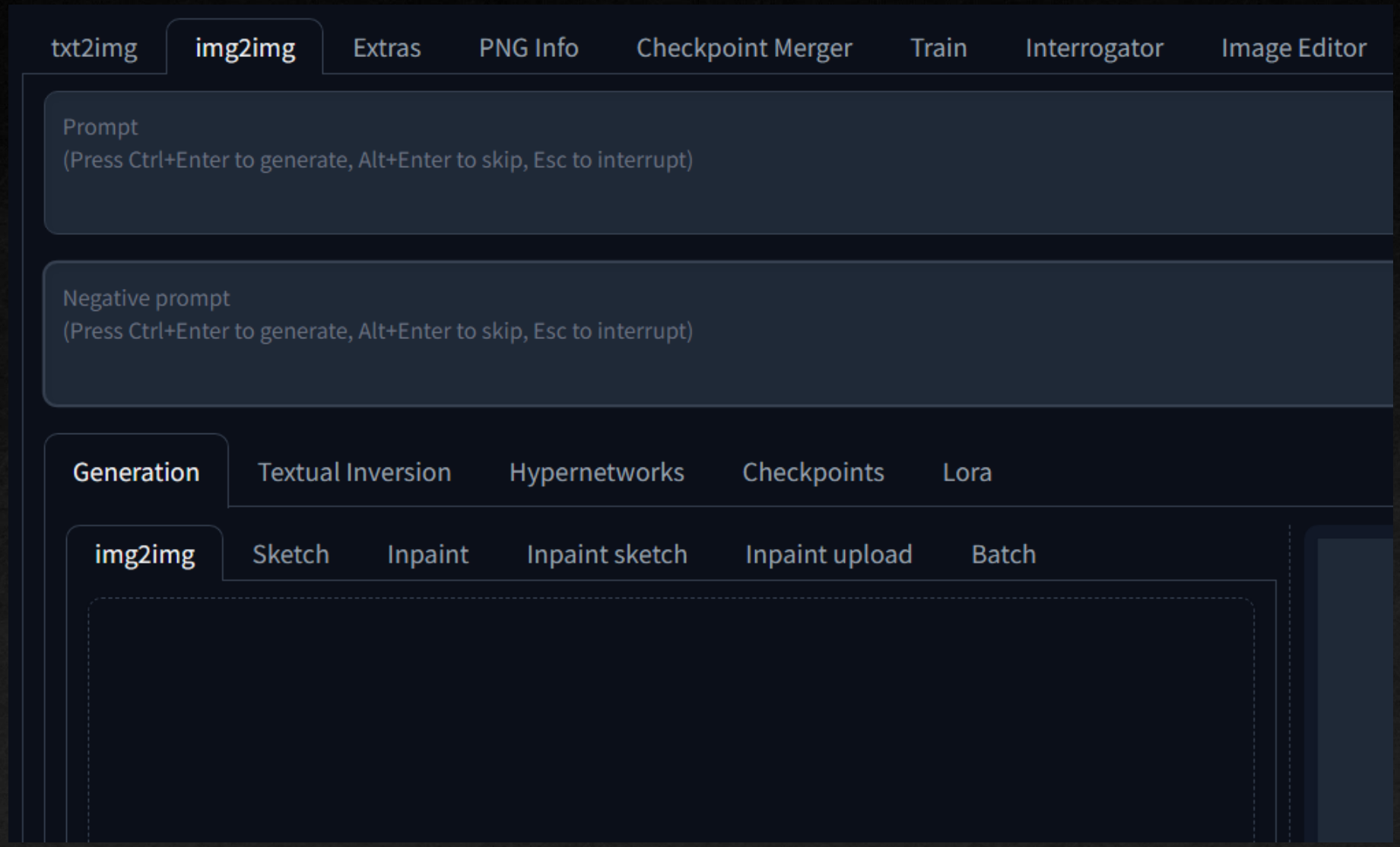




Edit your image with A.I.

Image to Image

การประมวลผล img2img



[1] Img2Img with Text

txt2img | img2img | Extras | PNG Info

1girl, yellow eyes, blonde hair, long hair, black tank top,

Resize to | Resize by

Scale 1

resize: from 816x1024 to 816x1024

Denoising strength 0.95





[2] Img2Img with Sketch

It helps the AI focus on your masked area.

[1] Select specific brush color

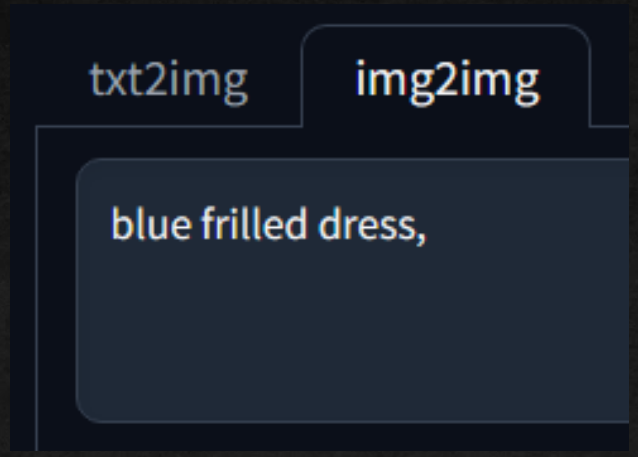
[2] Masking it....

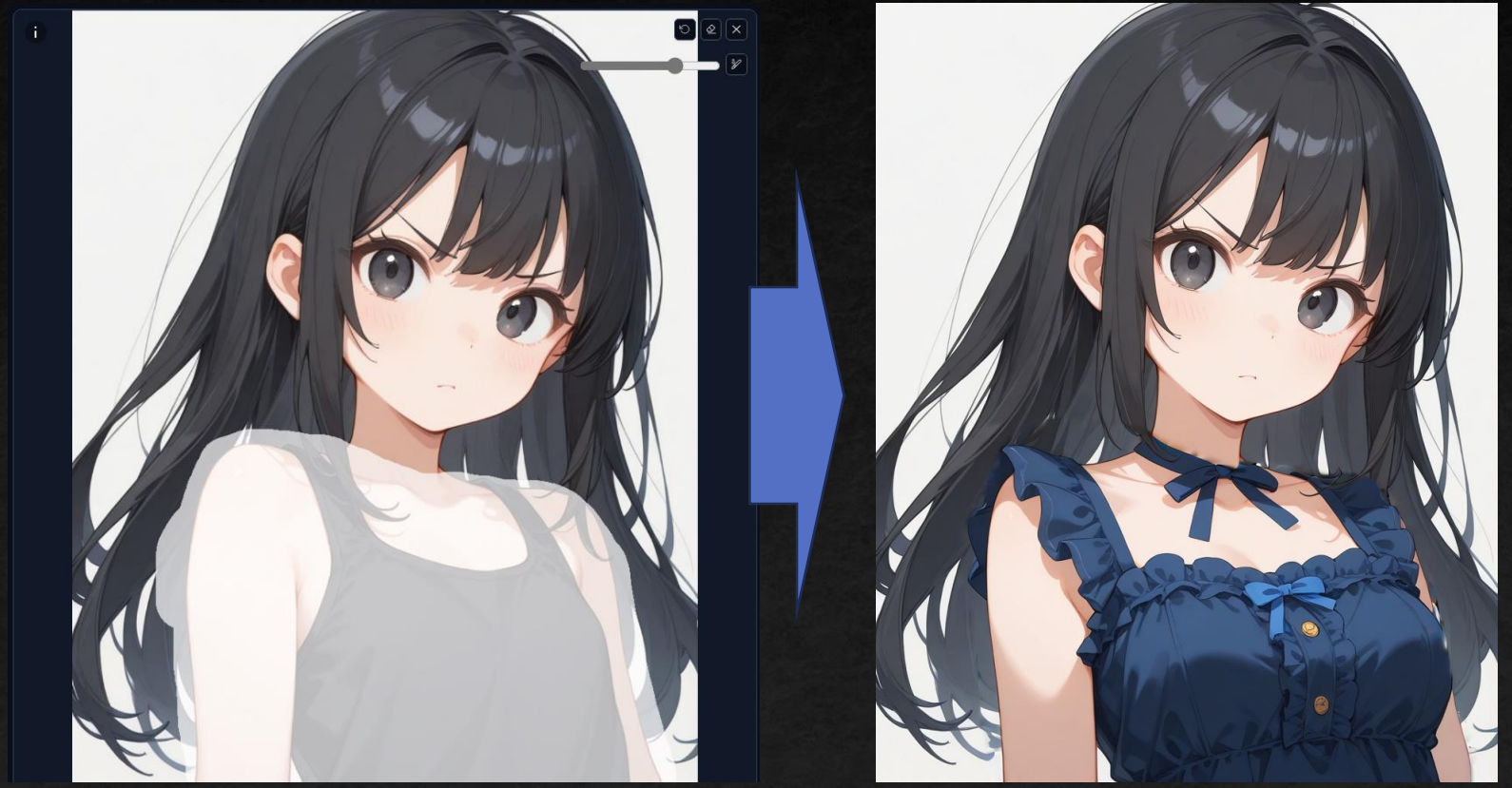
[3] Prompts & Generate



[3] Img2Img with Inpaint

This tool lets you add, remove, or enhance any part of the uploaded image.





[4] Img2Img with Inpaint-Sketch

it is a combination of the tools Sketch and Inpaint

You can mask any area of the uploaded image with a specific color, bringing your drawing to life on the image.

However, to use it, you should have proper painting knowledge



Learning by doing…